

Pratik Suvarna

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Objective

Detail-oriented Electronics and Telecommunication Engineering undergraduate with strong programming proficiency in Java, C++, and Python. Experienced in digital signal processing, embedded systems, and FPGA workflows. Seeking an internship in software development to contribute technical skills in a collaborative environment.

Education

- Sardar Patel Institute of Technology, Mumbai *[November 2022 – May 2026]*
[Bachelors of Technology in Electronics & Telecommunication Engineering]
Coursework: C++, C, Data Structures and Algorithms, Vitis, Java, Python
- PACE Junior College, Mumbai *[March 2020 – March 2022]*
[XII Standard with Electrical Maintenance as Bifocal]
- Thakur Public School *[March 2010 – March 2020]*
Coursework: Java, HTML

Skills

Languages: Java, C++, C, Python

Web Technologies: HTML

Database Management System: MySQL

Other: Matlab, Vitis, Vivado, GNU

Projects

1. Software Implementation of IC Circuit Boards:

- Designed and implemented an 8:1 Multiplexer using Vitis IDE, focusing on efficient digital logic design. The project was synthesized and deployed on the Edge Artix-7 FPGA board using Vivado. Functional testing was carried out to validate performance and correctness. This demonstrated proficiency in FPGA development and hardware description workflows.

2. Software Integration for Fetal ECG:

- Integrated fetal ECG digitally using Python for real-time data acquisition and Excel-based storage from data collected by an ESP-32 voltage reader.

3. Digital to Analog signal converter using Excel datasheet:

- Engineered a digital signal processing program in C++, leveraging Fast Fourier Transform algorithms to produce visual ECG graphs from Excel Datasheet, enabling cardiologists to analyse waveform morphology with 99.9% accuracy, which improved diagnostic process.

4. Digital Signal Processor in Java:

- Built a Java-based DSP tool supporting 6+ operations including FFT, filtering, modulation, and convolution. Achieved real-time processing on audio signals with dynamic time/frequency visualization.